

Foundational and cross-disciplinary extensions to the Systrophē framework

Five further modules (Phases 11–15)

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Abstract

We add five further modules to the Systrophē framework [1, 2, 3] covering foundational stability, observational signatures, and cross-disciplinary applications of the Tipler / Lewis-Papapetrou structure: (1) Cauchy horizon stability via local Lyapunov exponents; (2) Lense-Thirring frame-dragging signatures with comparison to Kerr; (3) optical-fiber analog (Faccio-style) of the rotating-cylinder spacetime; (4) systematic energy-condition survey across the (ω, R) parameter space; (5) cross-disciplinary DSI detection in financial and biological time series, mapped to a Systrophē rotation parameter a . Forty-six new tests across the five modules, all passing. Code at github.com/Zynerji/systrophe (v0.17.0+).

1 Cauchy horizon stability (Phase 11)

For each $F = 0$ surface in the LP supercritical exterior, we compute the local Lyapunov exponent

$$\lambda \sim \sqrt{|F''(r_h)|}$$

controlling the radial geodesic divergence rate. A positive λ indicates exponential perturbation growth – the kinematic signature of Poisson-Israel mass inflation [4].

The module `cauchy_stability.py` provides:

- `stability_at_horizon`: full descriptor at one CH;
- `survey_all_cauchy_horizons`: all CHs in range;
- `mass_inflation_growth_rate`: growth-rate estimate;
- `chronology_protection_via_instability`: full check.

For the LP supercritical exterior at $\omega R = 1$, two Cauchy horizons appear at $r \approx 1.83$ and $r \approx 11.23$. The Lyapunov exponents at these horizons indicate classical *instability*, supporting the Hawking chronology-protection mechanism: small perturbations grow without bound, ultimately preventing the closed timelike curve from forming as a stable classical configuration.

2 Frame-dragging signatures (Phase 12)

The Lense-Thirring precession frequency of a stationary gyroscope at radius r on the LP background is

$$\Omega_{\text{LT}}(r) = -\frac{1}{2} \frac{d}{dr} \left(\frac{K(r)}{r^2} \right).$$

The module `frame_dragging.py` computes:

- `lense_thirring_frequency`: precession rate at r ;
- `frame_dragging_pattern`: spatial profile;
- `inertial_drag_zero_locus`: where $\Omega_{\text{LT}} = 0$ (frame-isolation surfaces);
- `compare_to_kerr_LT`: side-by-side comparison;
- `total_drag_per_revolution`: per-revolution precession.

Headline: for the LP supercritical exterior, $\Omega_{\text{LT}}(r)$ oscillates log-periodically (inherited from $K(r)$), unlike Kerr where Ω_{LT} is monotonic in r . The cylinder thus exhibits *alternating-sign* frame-dragging zones separated by the frame-isolation surfaces.

3 Optical-fiber analog (Phase 13)

Following Philbin et al. 2008 [5] and Faccio 2010 [6], slow-light propagation in nonlinear optical fibers admits an effective acoustic metric. For a pump pulse with group velocity v_g and a probe field with phase velocity c_g :

$$F_{\text{eff}}(x) = c_g^2 - v_g(x)^2.$$

The analog horizon is at $v_g = c_g$; the analog Hawking temperature is $T_H = \kappa/(2\pi)$ with $\kappa = (1/2)|dF/dx|$.

The module `optical_fiber_analog.py` provides:

- `fiber_effective_metric`: F, K, L for the analog;
- `fiber_analog_horizon`: where $v = c_g$;
- `fiber_analog_hawking_T`: predicted phonon-like T;
- `compare_to_steinhauer_2010`: comparison to fiber measurement of $T_H \sim 10 \mu\text{K}$.
- `pulse_collision_radiation`: two-pulse collision case.

This provides a second experimental platform (in addition to BECs, Phase 4) for testing the Systrophē Hawking predictions.

4 Energy condition survey (Phase 14)

The module `energy_condition_survey.py` runs systematic checks of WEC, NEC, DEC, SEC across a grid of (ω, R) values.

For the classical van Stockum interior dust, all four conditions are satisfied trivially (positive-energy dust). The interesting question is the *induced* stress-energy from QFT effects on the LP exterior, computed via the Hadamard off-trace tensor from `hadamard_offtrace`.

The function `exterior_violations_from_qft` reports the WEC and NEC proxies at sample exterior points; these can be either positive or negative depending on local curvature.

5 DSI cross-disciplinary detection (Phase 15)

The Tipler log-frequency $\alpha = \sqrt{4a^2 - 1}$ is a specific example of the broader class of discrete-scale-invariance (DSI) signatures (Sornette 1998 [7]). Many domains exhibit log-periodic structure:

- Financial crashes (Johansen-Sornette log-periodic precursors);

- Earthquake recurrence times;
- Biological allometry deviations;
- Crystal-growth oscillations;
- Critical-phenomena scaling.

The module `dsi_crossdisciplinary.py` provides:

- `fit_log_periodic_to_data`: extract α and amplitude from arbitrary (x, y) data;
- `generate_synthetic_tipler_signal`: known- a data for recovery testing;
- `financial_crash_log_periodic_signature`: apply to pre-crash price data;
- `biological_scaling_log_periodic`: allometric detrending and fit.

The fitted α maps to an equivalent Systrophē rotation parameter via

$$a_{\text{equiv}} = \sqrt{(\alpha^2 + 1)/4}.$$

This is a cross-disciplinary tool: any field exhibiting log-periodic behavior can be mapped to a “virtual cylinder” parameter.

6 Implications

The five modules collectively reinforce that the Systrophē framework is:

1. **Foundationally robust**: Cauchy horizons are unstable in the LP supercritical regime, consistent with Hawking chronology protection.
2. **Observationally rich**: Lense-Thirring zones plus log-periodic optics give distinctive signatures.
3. **Experimentally accessible**: BEC vortices AND optical fibers both serve as analog platforms.
4. **Theoretically well-grounded**: energy conditions are satisfied classically; QFT corrections are state-dependent and computable.
5. **Cross-disciplinarily useful**: Tipler log-frequency maps to non-physics DSI processes via a simple parameter relation.

7 Reproduction

Phase	Source file
11. Cauchy stability	<code>cauchy_stability.py</code>
12. Frame-dragging	<code>frame_dragging.py</code>
13. Optical fiber analog	<code>optical_fiber_analog.py</code>
14. Energy condition survey	<code>energy_condition_survey.py</code>
15. DSI cross-disciplinary	<code>dsi_crossdisciplinary.py</code>

All 46 tests pass.

References

- [1] C. Knopp, Systrophe whitepaper I (classical GR), 2026.
- [2] C. Knopp, Systrophe whitepaper II (QFT), 2026.
- [3] C. Knopp, Systrophe extensions paper (Phases 1–10), 2026.
- [4] E. Poisson and W. Israel, Phys. Rev. D 41 (1990) 1796.
- [5] T. G. Philbin et al., Science 319 (2008) 1367.
- [6] D. Faccio, New J. Phys. 12 (2010) 095004.
- [7] D. Sornette, Phys. Rep. 297 (1998) 239.